DRV8313 TRIPLE HALF-H-BRIDGE DRIVER IC

Check for Samples: DRV8313

FEATURES

- **Three Half-H-Bridge Driver IC**
  - **Drives 3-Phase Brushless DC Motors**
  - **Individual Half-Bridge Control**
  - **Pins for Low-Side Current Sensing**
  - **Low MOSFET On-Resistance**
- **2.5-A Maximum Drive Current at 24 V, 25°C**
- **Uncommitted Comparator Can Be Used for Current Limit or Other Functions**
- **Built-In 3.3-V 10-mA LDO Regulator**
- **8-V to 60-V Operating Supply Voltage Range**
- **Thermally Enhanced Surface-Mount Package**

APPLICATIONS

- **HVAC Motors**
- **Consumer Products**
- **Office Automation Machines**
- **Factory Automation**
- **Robotics**

DESCRIPTION

The DRV8313 provides three individually controllable half-H-bridge drivers. It is intended to drive a three-phase brushless dc motor, though it can also be used to drive solenoids or other loads. Each output driver channel consists of N-channel power MOSFETs configured in a half-H-bridge configuration. The design brings the ground terminals of each driver to pins, to allow one to perform current sensing on each output.

Current-limit circuitry or other functions are possible uses of an uncommitted comparator.

The DRV8313 can supply up to 2.5-A peak or 1.75-A rms output current per channel (with proper PCB heatsinking at 24 V and 25°C) per half-H-bridge.

The device provides internal shutdown functions for overcurrent protection, short-circuit protection, undervoltage lockout, and overtemperature.

The DRV8313 comes in a 28-pin HTSSOP PowerPAD™ package.

ORDERING INFORMATION(1)

ORDERABLE PART NUMBER	PACKAGE(2)	TOPSIDE MARKING	SHIPPING
DRV8313PWPR	HTSSOP – PWP	DRV8313	Reel of 2000
DRV8313PWP			Tube of 50

(1) For the most-current packaging and ordering information, see the Package Option Addendum at the end of this document, or see the TI Web site at www.ti.com.

(2) See package drawings, thermal data, and symbolization at www.ti.com/packaging.

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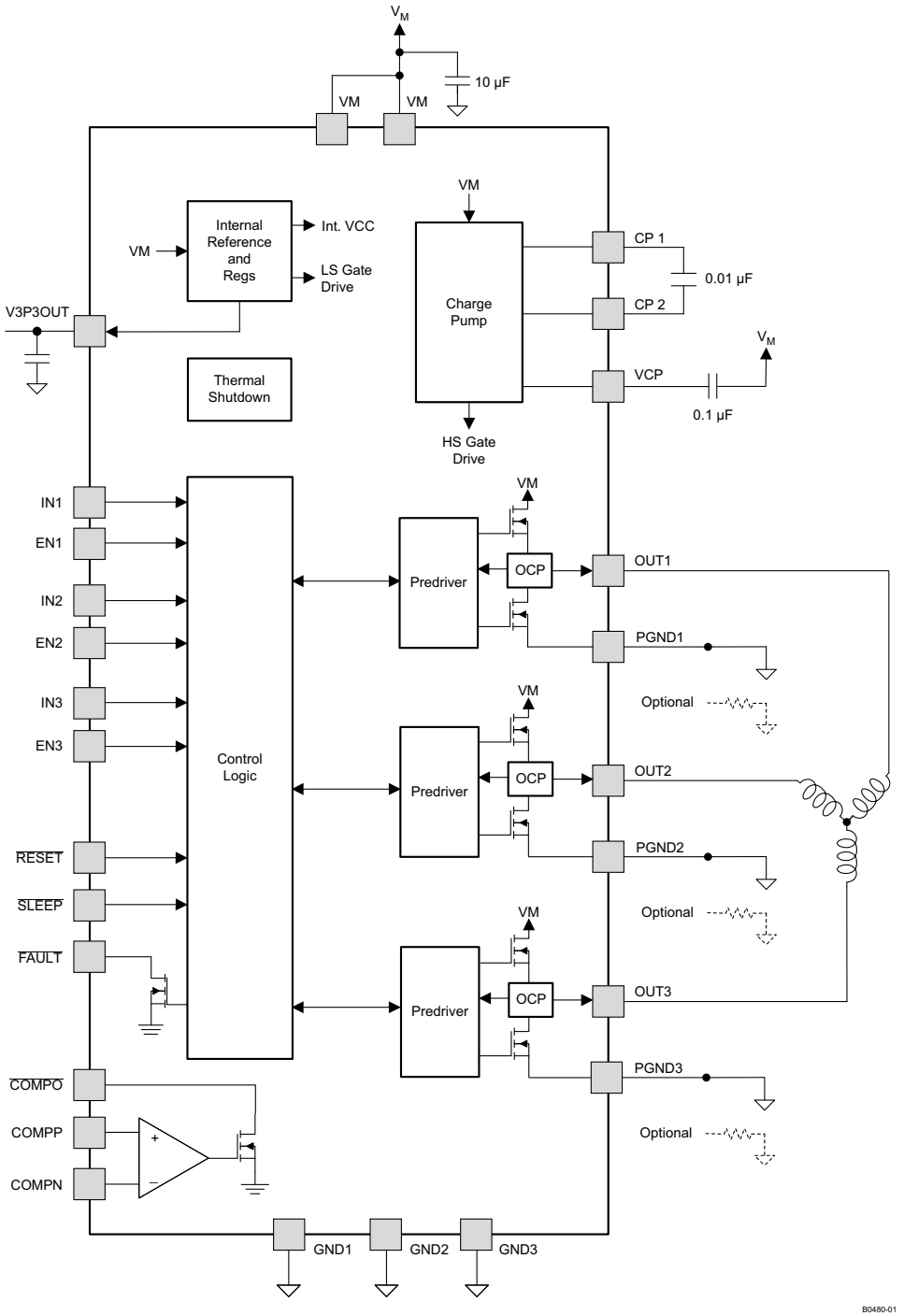
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This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

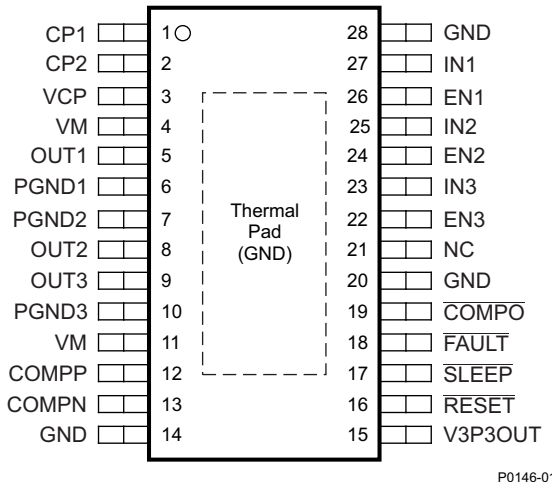
ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

FUNCTIONAL BLOCK DIAGRAM



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PWP Package
(Top View)



PIN DESCRIPTIONS

PIN		TYPE	DESCRIPTION	EXTERNAL COMPONENTS OR CONNECTIONS
NAME	NO.			
Power and Ground				
CP1	1	IO	Charge-pump flying capacitor	Connect a 0.01-µF 100-V capacitor between CP1 and CP2.
CP2	2	IO	Charge-pump flying capacitor	
GND	12, 20, 28, PPAD	–	Device ground	Connect to system ground
V3P3OUT	15	O	3.3-V regulator output	Bypass to GND with a 0.47-µF 6.3-V ceramic capacitor. Use for suppling external loads is permissible.
VCP	3	IO	High-side gate drive voltage	Connect a 0.1-µF 16-V ceramic capacitor to VM.
VM	4, 11	–	Main power supply	Connect to power supply (8.2 V–60 V). Connect both pins to the same supply. Bypass to GND with a 10-µF (minimum) capacitor.
Control				
EN1	26	I	Channel 1 enable	Logic high enables OUT1. Internal pulldown
EN2	24	I	Channel 2 enable	Logic high enables OUT2. Internal pulldown
EN3	22	I	Channel 3 enable	Logic high enables OUT3. Internal pulldown
IN1	27	I	Channel 1 input	Logic input controls state of OUT1. Internal pulldown
IN2	25	I	Channel 2 input	Logic input controls state of OUT2. Internal pulldown
IN3	23	I	Channel 3 input	Logic input controls state of OUT3. Internal pulldown
nRESET	16	I	Reset input	Active-low reset input initializes internal logic and disables the outputs. Internal pulldown
nSLEEP	17	I	Sleep-mode input	Logic high to enable device, logic low to enter low-power sleep mode. Internal pulldown
Status				
nFAULT	18	OD	Fault	Logic low when in fault condition (overtemperature, overcurrent, UVLO)
Comparator				
COMPN	13	I	Comparator negative input	Negative input of comparator
COMPP	12	I	Comparator positive input	Positive input of comparator
nCOMPO	19	OD	Comparator out	Output of comparator. Open-drain output

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PIN DESCRIPTIONS (continued)

PIN		TYPE	DESCRIPTION	EXTERNAL COMPONENTS OR CONNECTIONS
NAME	NO.			
Output				
OUT1	5	O	Output 1	Connect to loads.
OUT2	8	O	Output 2	
OUT3	9	O	Output 3	
PGND1	6	–	Ground for OUT1	Connect to ground, or to low-side current-sense resistors.
PGND2	7	–	Ground for OUT2	
PGND3	10	–	Ground for OUT3	

ABSOLUTE MAXIMUM RATINGS

over operating free-air temperature range (unless otherwise noted)(1)(2)

		VALUE	UNIT
	Power-supply voltage range (V_M)	–0.3 V to 65	V
	Digital-pin voltage range	–0.5 to 7	V
	Comparator input-voltage range	–0.5 to 7	V
	Peak motor-drive output current	Internally limited	A
	Pin voltage (GND1, GND2, GND3)	±600	mV
	Continuous motor-drive output current(3)	2.5	A
T_J	Operating virtual junction temperature range	–40 to 150	ºC
T_{stg}	Storage temperature range	–60 to 150	ºC

(1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions* is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
(2) All voltage values are with respect to the network ground terminal.
(3) Observe power dissipation and thermal limits.

THERMAL INFORMATION

THERMAL METRIC(1)		DRV8313 PWP 28 PINS	UNIT
θ_{JA}	Junction-to-ambient thermal resistance(2)	31.6	°C/W
θ_{JCtop}	Junction-to-case (top) thermal resistance(3)	15.9	°C/W
θ_{JB}	Junction-to-board thermal resistance(4)	5.6	°C/W
ψ_{JT}	Junction-to-top characterization parameter(5)	0.2	°C/W
ψ_{JB}	Junction-to-board characterization parameter(6)	5.5	°C/W
θ_{JCbot}	Junction-to-case (bottom) thermal resistance(7)	1.4	°C/W

(1) For more information about traditional and new thermal metrics, see the *IC Package Thermal Metrics* application report, SPRA953.
(2) The junction-to-ambient thermal resistance under natural convection is obtained in a simulation on a JEDEC-standard, high-K board, as specified in JESD51-7, in an environment described in JESD51-2a.
(3) The junction-to-case (top) thermal resistance is obtained by simulating a cold plate test on the package top. No specific JEDEC-standard test exists, but a close description can be found in the ANSI SEMI standard G30-88.
(4) The junction-to-board thermal resistance is obtained by simulating in an environment with a ring cold plate fixture to control the PCB temperature, as described in JESD51-8.
(5) The junction-to-top characterization parameter, ψ_{JT}, estimates the junction temperature of a device in a real system and is extracted from the simulation data for obtaining θ_{JA}, using a procedure described in JESD51-2a (sections 6 and 7).
(6) The junction-to-board characterization parameter, ψ_{JB}, estimates the junction temperature of a device in a real system and is extracted from the simulation data for obtaining θ_{JA} , using a procedure described in JESD51-2a (sections 6 and 7).
(7) The junction-to-case (bottom) thermal resistance is obtained by simulating a cold plate test on the exposed (power) pad. No specific JEDEC standard test exists, but a close description can be found in the ANSI SEMI standard G30-88.

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RECOMMENDED OPERATING CONDITIONS

over operating free-air temperature range (unless otherwise noted)

		MIN	NOM	MAX	UNIT
V_M	Motor power-supply voltage range (1)	8		60	V
V_{GNDX}	GND1, GND2, GND3 pin voltage	–500	0	500	mV
I_{V3P3}	V3P3OUT load current	0		10	mA

(1) All V_M pins must be connected to the same supply voltage.

ELECTRICAL CHARACTERISTICS

T_A = 25°C, over operating free-air temperature range (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
Power Supplies						
I_{VM}	VM operating supply current	V_M = 24 V, f_{PWM} < 50 kHz		1	5	mA
I_{VMQ}	VM sleep-mode supply current	V_M = 24 V		500	800	µA
V_{UVLO}	VM undervoltage lockout voltage	V_M rising		6.3	8	V
V3P3OUT Regulator						
V_{3P3}	V3P3OUT voltage	I_{OUT} = 0 to 10 mA	3.1	3.3	3.52	V
Logic-Level Inputs						
V_{IL}	Input low voltage			0.6	0.7	V
V_{IH}	Input high voltage		2.2		5.25	V
V_{HYS}	Input hysteresis		50		600	mV
I_{IL}	Input low current	VIN = 0	–5		5	µA
I_{IH}	Input high current	VIN = 3.3 V			100	µA
R_{PD}	Pulldown resistance			100		kΩ
nFAULT and COMPO OutputS (Open-Drain Outputs)						
V_{OL}	Output low voltage	I_O = 5 mA			0.5	V
I_{OH}	Output high leakage current	V_O = 3.3 V			1	µA
Comparator						
V_{CM}	Common-mode input-voltage range		0		5	V
V_{IO}	Input offset voltage		–7		7	mV
I_{IB}	Input bias current		–300		300	nA
t_R	Response time	100-mV step with 10-mV overdrive			2	µs
H-Bridge FETs						
$r_{ds(on)}$	High-side FET on-resistance	V_M = 24 V, I_O = 1 A, T_J = 25°C		0.24		Ω
		V_M = 24 V, I_O = 1 A, T_J = 85°C		0.29	0.39	
$r_{ds(on)}$	Low-side FET on-resistance	V_M = 24 V, I_O = 1 A, T_J = 25°C		0.24		Ω
		V_M = 24 V, I_O = 1 A, T_J = 85°C		0.29	0.39	
I_{OFF}	Off-state leakage current		–2		2	µA
t_{DEAD}	Output dead time			90		ns
Protection Circuits						
I_{OCP}	Overcurrent protection trip level		3			A
t_{OCP}	Overcurrent protection deglitch time			5		µs
T_{TSD}	Thermal shutdown temperature	Die temperature	150	160	180	°C

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SWITCHING CHARACTERISTICS(1)

T_A = 25°, V_M = 24 V, R_L = 20 Ω

NO.	PARAMETER	DESCRIPTION	MIN	MAX	UNIT
1	t_1	Delay time, ENx high to OUTx high, INx = 1	130	330	ns
2	t_2	Delay time, ENx low to OUTx low, INx = 1	275	475	ns
3	t_3	Delay time, ENx high to OUTx low, INx = 0	100	300	ns
4	t_4	Delay time, ENx low to OUTx high, INx = 0	200	400	ns
5	t_5	Delay time, INx high to OUTx high	300	500	ns
6	t_6	Delay time, INx low to OUTx low	275	475	ns
7	t_r	Output rise time, resistive load to GND	30	150	ns
8	t_f	Output fall time, resistive load to GND	30	150	ns

(1) Not production tested

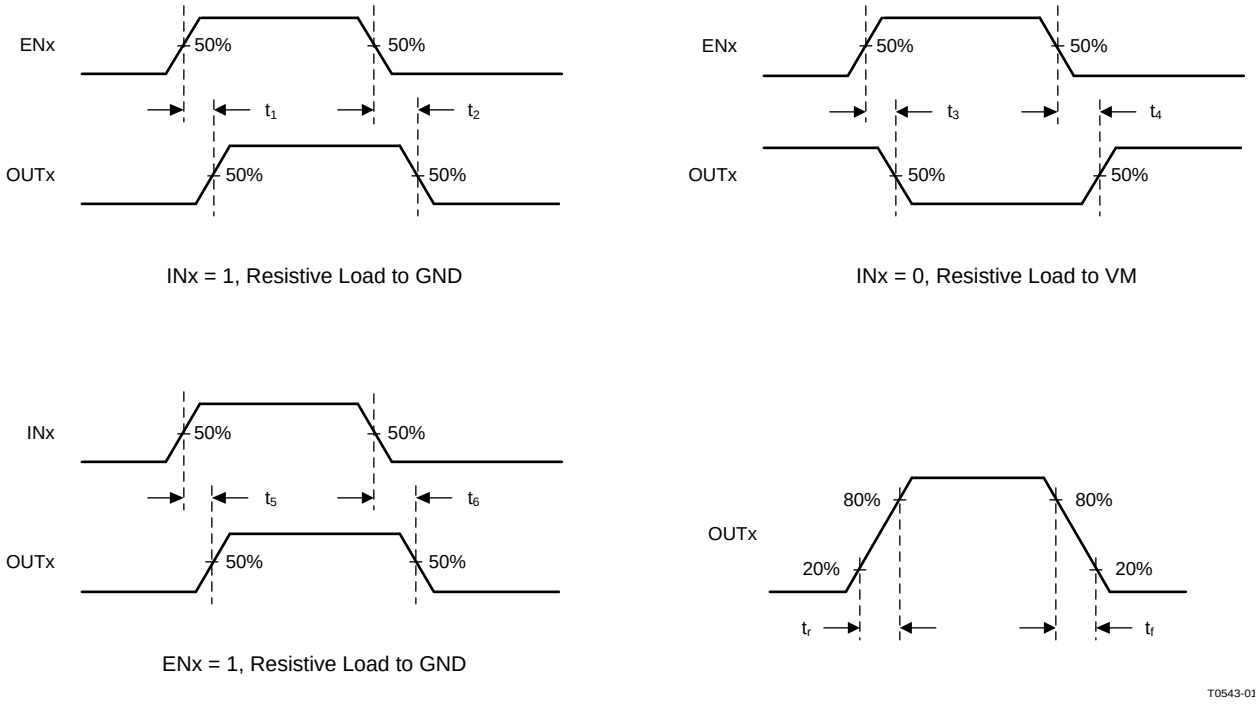


Figure 1. DRV8313 Switching Characteristics

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FUNCTIONAL DESCRIPTION

Output Stage

The DRV8313 contains three half-H-bridge drivers. The source terminals of the low-side FETs of all three half-H-bridges terminate at separate pins (GND1, GND2, and GND3) to allow the use of a low-side current-sense resistor on each output, if desired. The user may also connect all three together to a single low-side sense resistor, or may connect them directly to ground if there is no need for current sensing.

If using a low-side sense resistor, take care to ensure that the voltage on the GND1, GND2, or GND3 pin does not exceed ±500 mV.

Note that there are multiple VM motor power-supply pins. Connect all VM pins together to the motor-supply voltage.

Bridge Control

The INx input pins directly control the state (high or low) of the OUTx outputs; the ENx input pins enable or disable the OUTx driver. The following table shows the logic:

INx	ENx	OUTx
X	0	Z
0	1	L
1	1	H

Charge Pump

Because the output stages use N-channel FETs, the device requires a gate-drive voltage higher than the VM power supply to enhance the high-side FETs fully. The DRV8313 integrates a charge-pump circuit that generates a voltage above the VM supply for this purpose.

The charge pump requires two external capacitors for operation. See the block diagram and pin descriptions for details on these capacitors (value, connection, and so forth).

The charge pump shuts down when nSLEEP is active-low.

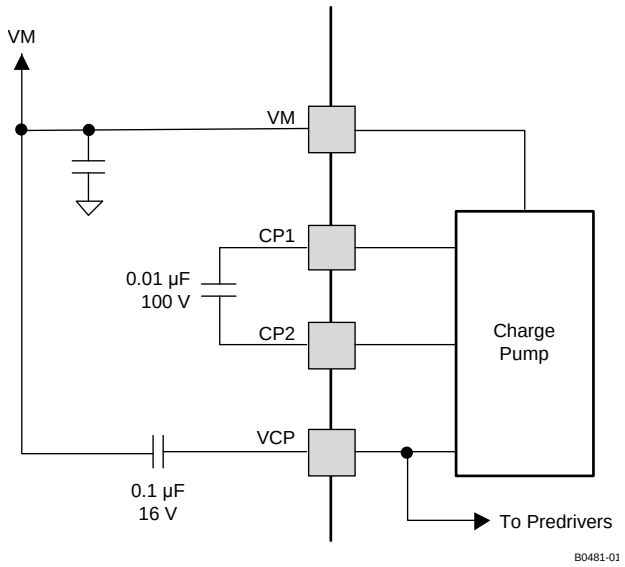


Figure 2. DRV8313 Charge Pump

Comparator

The DRV8313 includes an uncommitted comparator, which can find use as a current-limit comparator or for other purposes.

The following diagram shows connections to use the comparator to sense current for implementing a current limit. Current from all three low-side FETs is sensed using a single low-side sense resistor. The voltage across the sense resistor is compared with a reference, and when the sensed voltage exceeds the reference, a current-limit condition is signaled to the controller. The V3P3OUT internal voltage regulator can be used to set the reference voltage of the comparator.

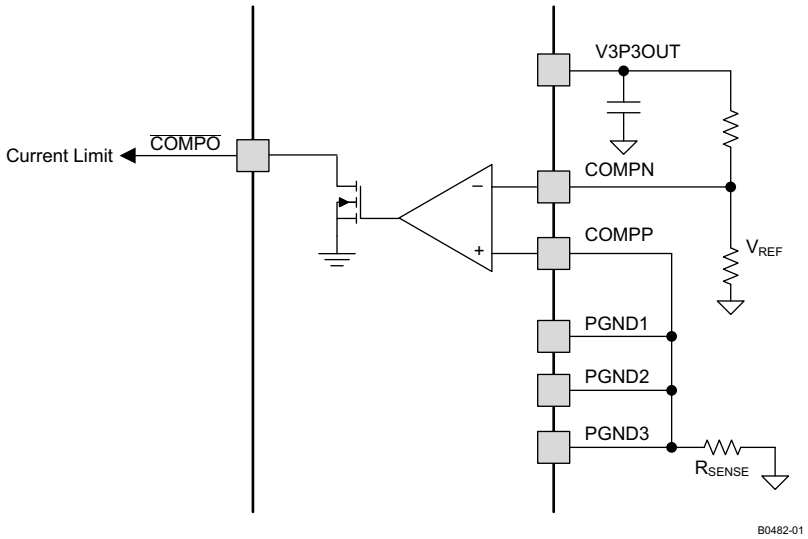


Figure 3. DRV8313 Comparator

nRESET and nSLEEP Operation

The nRESET pin, when driven active-low, resets any faults. It also disables the output drivers while it is active. The device ignores all inputs while nRESET is active. Note that there is an internal power-up-reset circuit, so that driving nRESET at power up is not required.

Driving nSLEEP low puts the device into a low-power sleep state. Entering this state disables the output drivers, stops the gate-drive charge pump, resets all internal logic (including faults), and stops all internal clocks. In this state, the device ignores all inputs until nSLEEP returns inactive-high. When returning from sleep mode, some time (approximately 1 ms) must pass before the motor driver becomes fully operational. Note that the V3P3 regulator remains operational in sleep mode.

Protection Circuits

The DRV8313 has full protection against undervoltage, overcurrent, and overtemperature events.

OVERCURRENT PROTECTION (OCP)

An analog current-limit circuit on each FET limits the current through the FET by removing the gate drive. If this analog current limit persists for longer than the OCP deglitch time, the device disables the channel experiencing the overcurrent and drives the nFAULT pin low. The driver remains off until either assertion of nRESET or the cycling of VM power.

Overcurrent conditions on both high- and low-side devices, that is, a short to ground, supply, or across the motor winding, all result in an overcurrent shutdown.

THERMAL SHUTDOWN (TSD)

If the die temperature exceeds safe limits, the device disables all outputs and drives the nFAULT pin low. Once the die temperature has fallen to a safe level, operation automatically resumes.

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UNDERVOLTAGE LOCKOUT (UVLO)

If at any time the voltage on the VM pins falls below the undervoltage-lockout threshold voltage, the device disables all outputs, resets internal logic, and drives the nFAULT pin low. Operation resumes when VM rises above the UVLO threshold.

THERMAL INFORMATION

Thermal Protection

The DRV8313 has thermal shutdown (TSD) as previously described. A die temperature in excess of approximately 150°C disables the device until the temperature drops to a safe level.

Any tendency of the device to enter thermal shutdown is an indication of excessive power dissipation, insufficient heatsinking, or too high an ambient temperature.

Power Dissipation

The power dissipated in the output FET resistance, or $r_{DS(on)}$ dominates power dissipation in the DRV8313. A rough estimate of average power dissipation of each half-H-bridge when running a static load is:

$$P = r_{DS(on)} \times (I_{OUT})^2 \quad (1)$$

where P is the power dissipation of one H-bridge, $r_{DS(on)}$ is the resistance of each FET, and I_{OUT} is equal to the average current drawn by the load. Note that at start-up and fault conditions, this current is much higher than normal running current; remember to take these peak currents and their duration into consideration.

The total device dissipation is the power dissipated in each of the three half-H-bridges added together.

The maximum amount of power that the device can dissipate depends on ambient temperature and heatsinking.

Note that $r_{DS(on)}$ increases with temperature, so as the device heats, the power dissipation increases. Take this into consideration when sizing the heatsink.

Heatsinking

The PowerPAD package uses an exposed pad to remove heat from the device. For proper operation, this pad must be thermally connected to copper on the PCB to dissipate heat. On a multi-layer PCB with a ground plane, add a number of vias to connect the thermal pad to the ground plane to accomplish this. On PCBs without internal planes, add copper area on either side of the PCB to dissipate heat. If the copper area is on the opposite side of the PCB from the device, use thermal vias to transfer the heat between the top and bottom layers.

For details about how to design the PCB, see TI Application Report SLMA002, *PowerPAD Thermally Enhanced Package* and TI Application Brief SLMA004, *PowerPAD Made Easy*, available at www.ti.com.

In general, providing more copper area allows the dissipation of more power.

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APPLICATION INFORMATION

Output Configurations and Connections

The typical application for the DRV8313 is to drive a 3-phase brushless motor. In this application, the three outputs connect to the three motor leads, as shown in Figure 4.

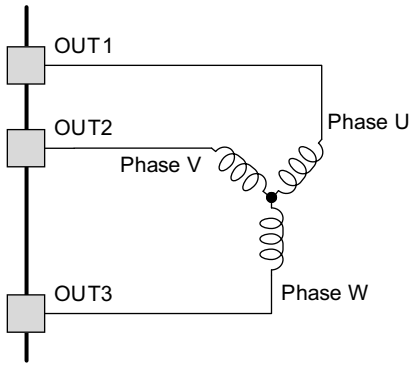


Figure 4. Three-Phase Motor Connection

The device achieves standard 120° (also called trapezoidal or block) commutation, using synchronous rectification, by following the states shown in Table 1

Table 1. Three-Phase Motor Signals

State	OUT1 (Phase U)			OUT2 (Phase V)			OUT3 (Phase W)		
	IN1	EN1	OUT1	IN2	EN2	OUT2	IN3	EN3	OUT3
1	X	0	Z	1 / PWM	1	H / PWM	0	1	L
2	1 / PWM	1	H / PWM	X	0	Z	0	1	L
3	1 / PWM	1	H / PWM	0	1	L	X	0	Z
4	X	0	Z	0	1	L	1 / PWM	1	H / PWM
5	0	1	L	X	0	Z	1 / PWM	1	H / PWM
6	0	1	L	1 / PWM	1	H / PWM	X	0	Z

On can implement asynchronous rectification by also applying the PWM signal to the enable inputs.

The DRV8313 can drive other loads, including dc brush motors and solenoids. For example, one could drive a dc brush motor in both directions, plus a single solenoid or unidirectional dc brush motor:

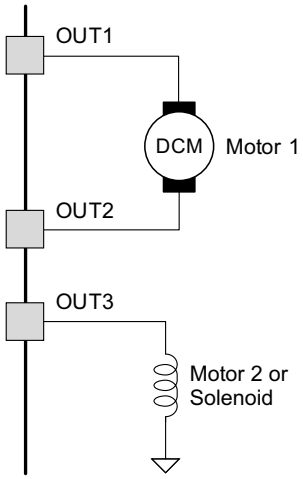


Figure 5. Bidirectional Motor Plus Motor or Solenoid Connection

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The functions would be as shown in Table 2.

Table 2. Bidirectional Motor Plus Motor or Solenoid Signals

Motor 1							Motor 2 or Solenoid			
Function	**IN1**	**EN1**	**OUT1**	**IN2**	**EN2**	**OUT2**	**Function**	**IN3**	**EN3**	**OUT3**
Off or coast	X	0	Z	X	X	X	On	1 / PWM	1	1
Off or coast	X	X	X	X	0	X	Off or slow decay	0	1	0
Forward	1 / PWM	1	1	0	1	0	Off or coast	X	0	X
Reverse	0	1	0	1 / PWM	1	1				
Brake or slow decay	0	1	0	0	1	0				
Brake or slow decay	1	1	1	1	1	1				

Applying a PWM signal to the appropriate INx pin(s) as shown in Table 2 could implement PWM speed control.

Another possibility is controlling three different loads. Note that it is possible to return one side of the load either to the power supply (VM) or to ground.

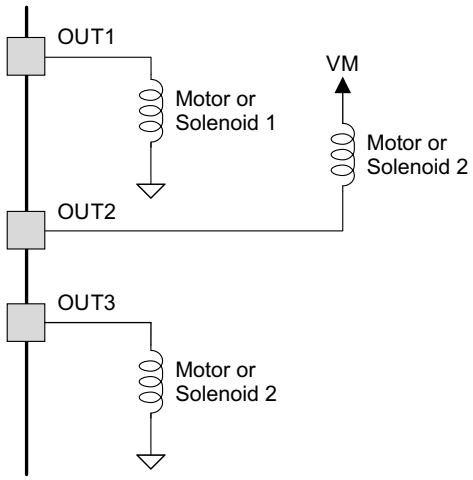


Figure 6. Three Independent Load Connections

Table 3. Three Independent Load Signals

Motor or Solenoid 1				Motor or Solenoid 2				Motor or Solenoid 3			
Function	**IN1**	**EN1**	**OUT1**	**Function**	**IN2**	**EN2**	**OUT2**	**Function**	**IN3**	**EN3**	**OUT3**
On	1 / PWM	1	1	On	1 / PWM	1	1	On	1 / PWM	1	1
Off or slow decay	0	1	0	Off or slow decay	0	1	0	Off or slow decay	0	1	0
Off or coast	X	0	X	Off or coast	X	0	X	Off or coast	X	0	X

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PACKAGING INFORMATION

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead/Ball Finish	MSL Peak Temp (3)	Samples (Requires Login)
DRV8313PWP	ACTIVE	HTSSOP	PWP	28	50	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-3-260C-168 HR	
DRV8313PWPR	ACTIVE	HTSSOP	PWP	28	2000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-3-260C-168 HR	

(1) The marketing status values are defined as follows:
ACTIVE: Product device recommended for new designs.
LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.
NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.
PREVIEW: Device has been announced but is not in production. Samples may or may not be available.
OBSOLETE: TI has discontinued the production of the device.

(2) Eco Plan - The planned eco-friendly classification: Pb-Free (RoHS), Pb-Free (RoHS Exempt), or Green (RoHS & no Sb/Br) - please check http://www.ti.com/productcontent for the latest availability information and additional product content details.
TBD: The Pb-Free/Green conversion plan has not been defined.
Pb-Free (RoHS): TI's terms "Lead-Free" or "Pb-Free" mean semiconductor products that are compatible with the current RoHS requirements for all 6 substances, including the requirement that lead not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, TI Pb-Free products are suitable for use in specified lead-free processes.
Pb-Free (RoHS Exempt): This component has a RoHS exemption for either 1) lead-based flip-chip solder bumps used between the die and package, or 2) lead-based die adhesive used between the die and leadframe. The component is otherwise considered Pb-Free (RoHS compatible) as defined above.
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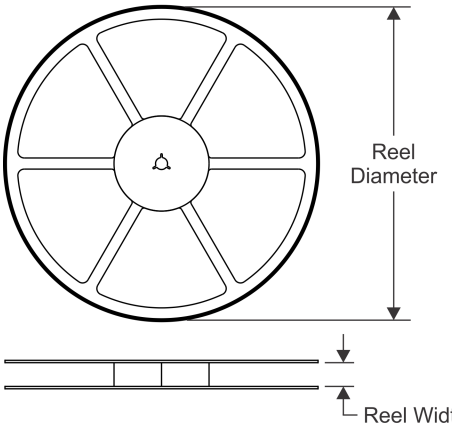
(3) MSL, Peak Temp. -- The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

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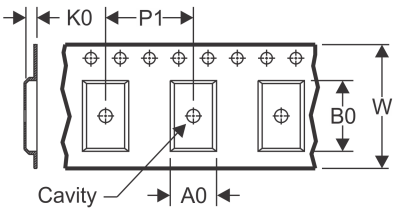
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TAPE AND REEL INFORMATION

REEL DIMENSIONS

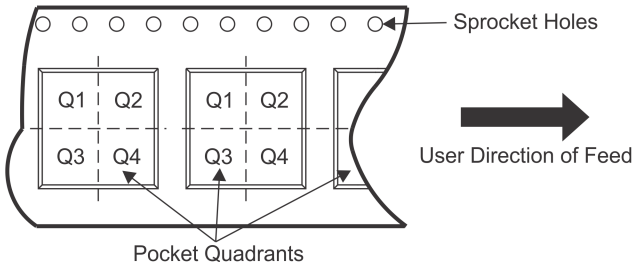


TAPE DIMENSIONS



A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

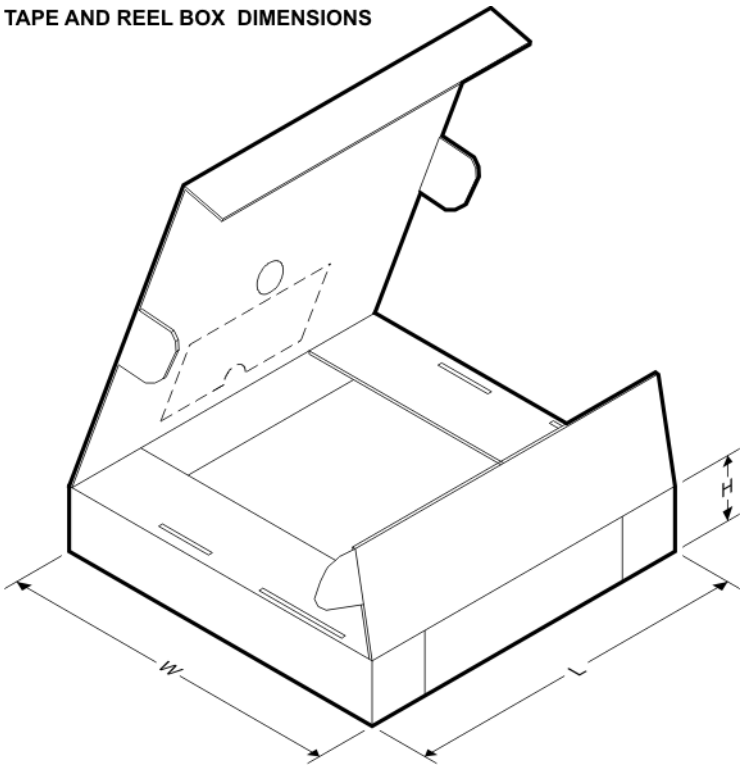
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
DRV8313PWPR	HTSSOP	PWP	28	2000	330.0	16.4	6.9	10.2	1.8	12.0	16.0	Q1

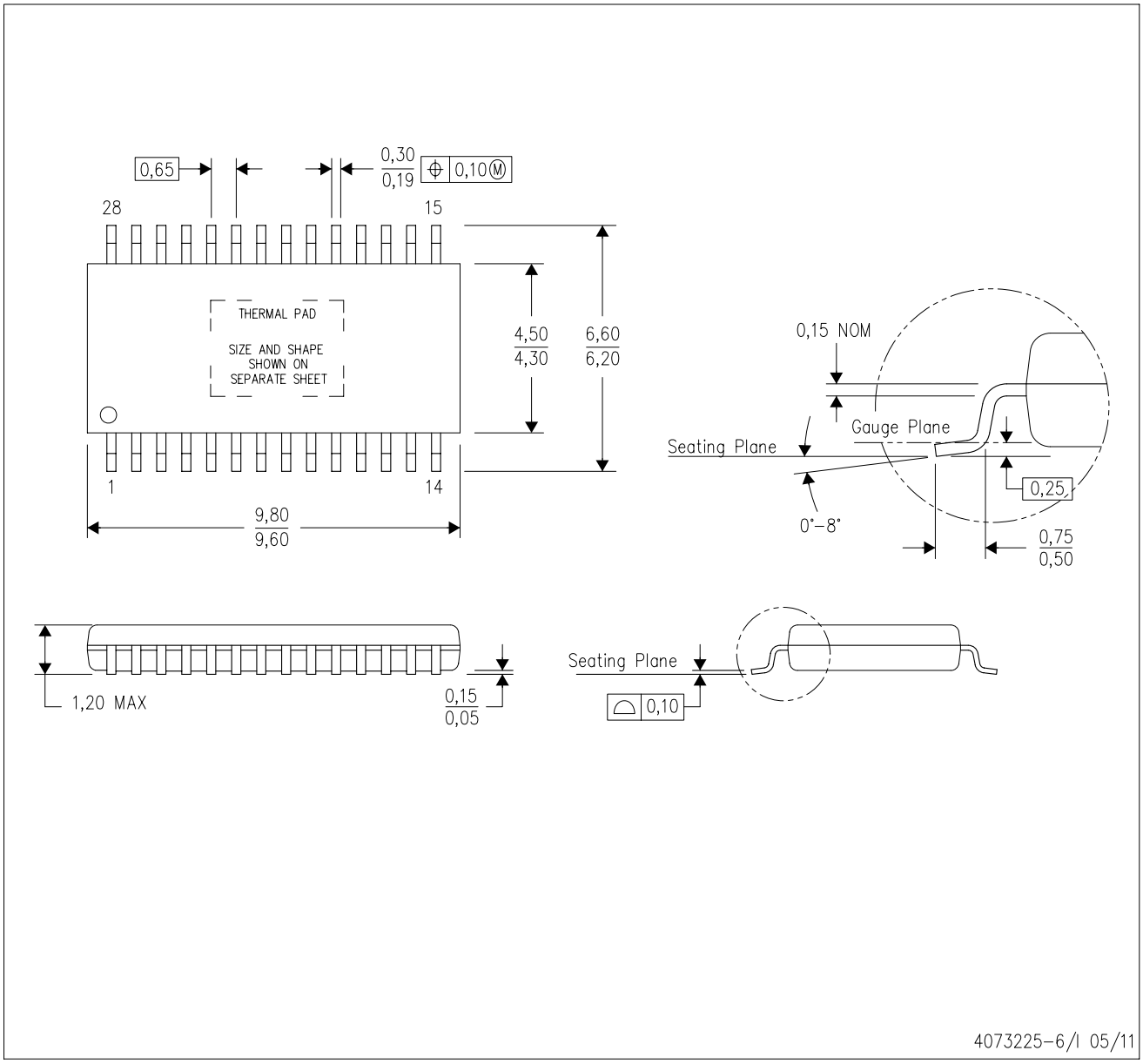
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
DRV8313PWPR	HTSSOP	PWP	28	2000	367.0	367.0	38.0

PWP (R–PDSO–G28) PowerPAD™ PLASTIC SMALL OUTLINE



NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Body dimensions do not include mold flash or protrusions. Mold flash and protrusion shall not exceed 0.15 per side.
D. This package is designed to be soldered to a thermal pad on the board. Refer to Technical Brief, PowerPad Thermally Enhanced Package, Texas Instruments Literature No. SLMA002 for information regarding recommended board layout. This document is available at www.ti.com <http://www.ti.com>.
E. See the additional figure in the Product Data Sheet for details regarding the exposed thermal pad features and dimensions.
E. Falls within JEDEC MO–153

PowerPAD is a trademark of Texas Instruments.

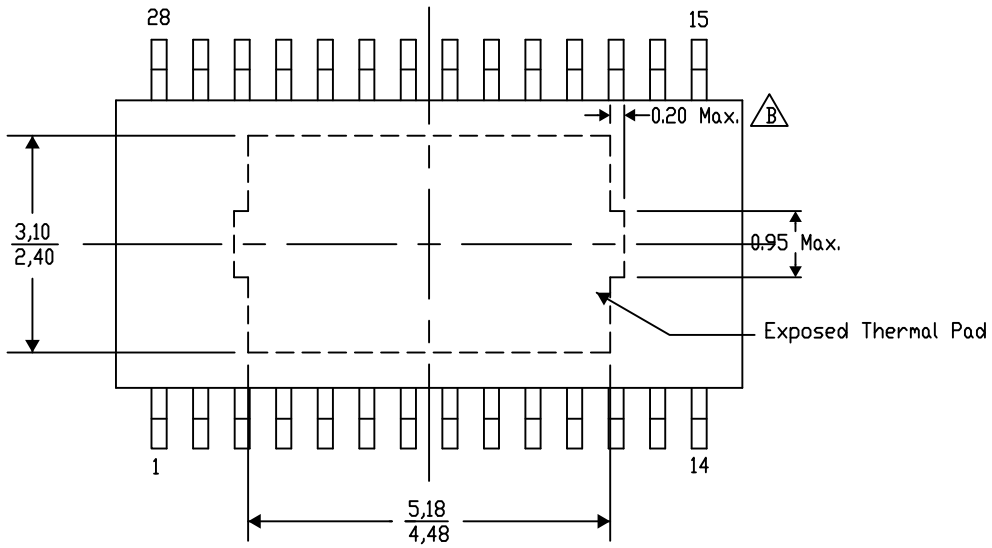
PWP (R–PDSO–G28) PowerPAD™ SMALL PLASTIC OUTLINE

THERMAL INFORMATION

This PowerPAD™ package incorporates an exposed thermal pad that is designed to be attached to a printed circuit board (PCB). The thermal pad must be soldered directly to the PCB. After soldering, the PCB can be used as a heatsink. In addition, through the use of thermal vias, the thermal pad can be attached directly to the appropriate copper plane shown in the electrical schematic for the device, or alternatively, can be attached to a special heatsink structure designed into the PCB. This design optimizes the heat transfer from the integrated circuit (IC).

For additional information on the PowerPAD package and how to take advantage of its heat dissipating abilities, refer to Technical Brief, PowerPAD Thermally Enhanced Package, Texas Instruments Literature No. SLMA002 and Application Brief, PowerPAD Made Easy, Texas Instruments Literature No. SLMA004. Both documents are available at www.ti.com.

The exposed thermal pad dimensions for this package are shown in the following illustration.



Top View

Exposed Thermal Pad Dimensions

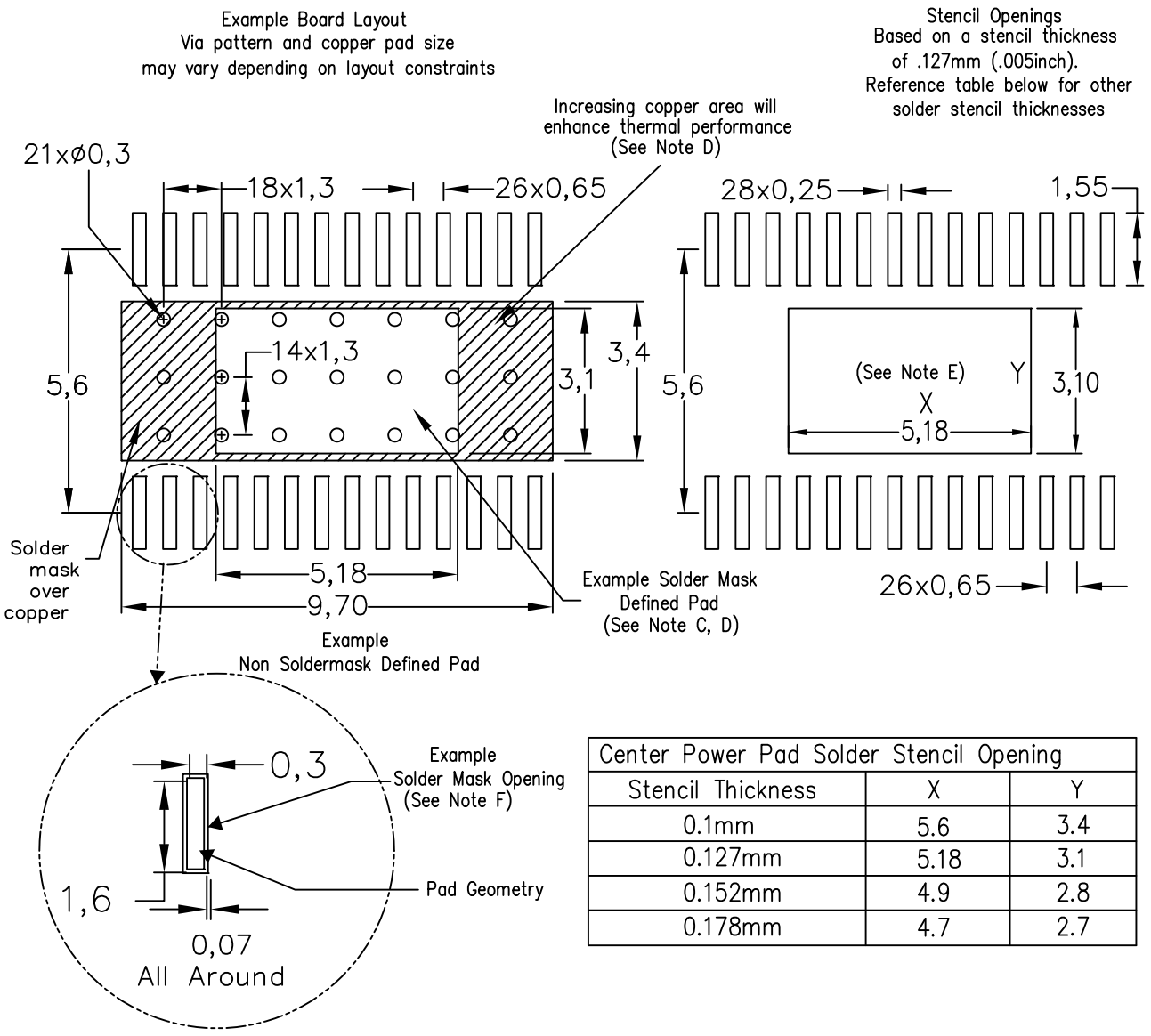
4206332–38/AC 07/12

NOTE: A. All linear dimensions are in millimeters

B. Exposed tie strap features may not be present.

PowerPAD is a trademark of Texas Instruments

PWP (R–PDSO–G28) PowerPAD™ PLASTIC SMALL OUTLINE



Center Power Pad Solder Stencil Opening		
Stencil Thickness	X	Y
0.1mm	5.6	3.4
0.127mm	5.18	3.1
0.152mm	4.9	2.8
0.178mm	4.7	2.7

4207609–22/R 07/12

NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Customers should place a note on the circuit board fabrication drawing not to alter the center solder mask defined pad.
D. This package is designed to be soldered to a thermal pad on the board. Refer to Technical Brief, PowerPad Thermally Enhanced Package, Texas Instruments Literature No. SLMA002, SLMA004, and also the Product Data Sheets.
E. For specific thermal information, via requirements, and recommended board layout. These documents are available at www.ti.com <http://www.ti.com>. Publication IPC–7351 is recommended for alternate designs. Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Example stencil design based on a 50% volumetric metal load solder paste. Refer to IPC–7525 for other stencil
F. Customers should contact their board fabrication site for solder mask tolerances between and around signal pads.

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